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Urban spatial structure, suburbanization and transportation in Barcelona ☆

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ABSTRACT

I investigate the effect of improvements to the transportation infrastructure on changes in location patterns of population in Barcelona, Spain between 1991 and 2006. At a census tract level, I verify and extend the finding of Baum-Snow (2007a) that transportation improvements cause suburbanization: (1) improvements to the highway and railroad systems foster population growth in suburban areas; (2) the transit system also affects the location of population inside the central business district (CBD). To estimate the causal relationship between the growth of population (density) and transportation improvements, I rely on an instrumental variables estimation that uses distances to the nearest Roman road, the nearest 19th century main road, and the nearest 19th century railroad network as instruments for the 1991–2001 changes in distance to the nearest highway ramp and the distance to the nearest railroad station.

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1. Introduction

I investigate the effects of highway and railroad improvements on changes in urban spatial structure that occurred in the Barcelona Metropolitan Region (BMR) between 1991 and 2006. I find that improvements to the transportation infrastructure cause BMR suburbanization and influence its spatial pattern by attracting population to non-central suburban tracts that in 1991–2001 had improved access to the highway system and to central and non-central suburban tracts near railroad stations. Besides causing suburbanization, transportation also affects residential location decisions in the CBD by attracting population near railroad stations. Overall, these results confirm that transportation infrastructure and its improvements are important elements needed to explain urban spatial structure in general and the process and spatial pattern of suburbanization in particular.

This investigation is of interest for three reasons. First, it furthers our understanding of the role of transportation infrastructure in shaping cities. Baum-Snow (2007a) shows that transportation improvements cause suburbanization. At a census tract level and for different city areas, my results confirm this finding and indicate that improvements to the transportation infrastructure also influence the spatial pattern of suburbanization. Since suburbanization leads to greater resource consumption and CO_2 emissions (Kahn, 2000), and to an inefficient supply of public services (Bertaud, 2002), the findings reported here provide a basis for analyzing potential policy interventions that seek to redirect the spatial pattern of suburbanization and mitigate its negative consequences.

Second, these new results are also useful for urban planners in two additional ways. Since improvements in transportation infrastructure cause changes in population location patterns, planning policies should also consider the complementary changes in the spatial distribution of the demand for public services. Furthermore, since the effect of transportation improvements depends on the city area where infrastructure is located, planning policies should be spatially differentiated.

Finally, this research is important because it is centered on a European city and thus permits comparison with previous studies of US cities. Despite differences in city population size and density, land-use planning, or the use of public transit between US and European cities, these results show that suburbanization is an ongoing phenomenon in a big European city and that it is also related to the transportation infrastructure.

The empirical strategy that I develop below is based on estimating the relationship between the growth of population (density) in

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a census tract and its transportation improvements. Such improvements include the changes in census tract distance to the nearest highway ramp and the census tract distance to the nearest railroad station. Departing from this unconditional relationship, I gradually control for the initial level of population density, proximity to the main centers, geography, and history (past populations). I implement this strategy for three different city areas: CBD, subcenters and non-central suburban areas.

To carefully establish this causal relationship, I consider an important identification problem: the simultaneous determination of population growth and transportation improvements. Planners may wish to serve areas expected to have strong population growth or, alternatively, they may choose to serve areas expected to have weak growth. In both cases, reverse causation would be at work. To solve the identification problem I use instruments as sources of exogenous variation for transportation improvements.

Based on the history of the infrastructure development in Barcelona (and in Spain more broadly), I derive three instruments from maps of the Roman roads, main roads constructed before the end of the 19th century, and the railroad lines built during this same time period. These old infrastructures were not placed randomly throughout the landscape. On the contrary, their supply and location was influenced by the size of contemporary populations and the suitability of geography for building these infrastructures. Moreover, these factors may have also influenced modern transportation improvements. Thus the abovementioned variables of geography and history (past populations) are required in order to guarantee the exogeneity of these instruments.

This research contributes to the existing literature in two ways. First, it extends the conclusions of Baum-Snow (2007a) who investigated the effect of the interstate highway system on suburbanization and found that highway improvements contribute to central city population decline. Using data on a much finer geographic scale, I also investigate the effect of transportation improvements inside the central city and the suburbs (suburban subcenters and non-central suburban areas). I also distinguish between the effects of the highway system and the railroad network. Results reported here thus suggest that transportation improvements influence the suburbanization process by attracting population near railroad stations and to non-central suburban areas that improve access to the highway system.

The present work is also related to the large literature on urban spatial structure. Despite the central role of transportation infrastructure in theoretical models of urban spatial structure, much of the empirical literature is concerned with its effect on the spatial distribution of land prices. Only a small number of papers study the relationship between transportation and the spatial distribution of population and employment.[1] With the exception of Baum-Snow (2007a), previous work has not considered the simultaneous determination of population density (growth) and transportation infrastructure (improvements). As mentioned above, I resolve this endogeneity problem by exploiting the exogenous variation in three instrumental variables.

The remainder of the paper is structured as follows. In the next section, I propose an empirical strategy for estimating the effects of transportation infrastructure improvements on population (density) growth. In Section 3, I describe transportation infrastructure of Barcelona and I then present the main results in Section 4. Finally I present conclusions in Section 5.

[1] Steen (1986), Baum-Snow (2007a), and Garcia-López (2010) estimate the effect of transportation infrastructure on population density patterns, and McMillen and McDonald (1998) on employment density patterns. Boarnet (1994a,b) and Bollinger and Ihlanfeldt (1997, 2003) consider transportation infrastructure (improvements) as a possible determinant of intrametropolitan population and employment growth.

2. Urban spatial structure and transportation: theory and estimation

The classical monocentric land use theory developed by Alonso (1964), Mills (1967) and Muth (1969) (AMM) considers a non-limited, radial-type transportation infrastructure covering the whole city in the same way and therefore allowing the same access to the unique main center or CBD from any point located at the same distance from this CBD. The AMM way of modeling transportation infrastructure leads to a homogeneous reduction in population density as population moves away from the CBD (the CBD gradient), but not as population moves away from transportation infrastructure. When improvements in transportation infrastructure increase transport speed, population density decreases near the CBD and increases in the suburbs and, as a result, the CBD gradient flattens (Wheaton, 1974).[2]

To motivate my econometric strategy, the works by Anas and Moses (1979) and Baum-Snow (2007b) are more useful because they deal with transportation infrastructure in a more realistic way. Anas and Moses (1979) extend the AMM model by considering two competing transportation infrastructures. First, the classical AMM-transportation infrastructure based on a dense network of radial streets. Second, a high speed transit system based on sparse radial corridors. Depending on the cost of alternative transportation modes, the authors show that both rents and population densities might decrease with distance to the transit lines (a transportation infrastructure gradient).

More recently, Baum-Snow (2007b) incorporates highways into the transportation infrastructure of a monocentric city. Population commutes to the CBD using these faster sparse radial highways or using a dense network of streets that connects each point in the city to the CBD and to the nearest highway.[3] Concerned with the suburbanization process and the impact of highway improvements previously documented in Baum-Snow (2007a), the author shows that new highways affect urban spatial structure by causing population to spread out along the highways and, as a result, by increasing population densities in areas near highways and decreasing elsewhere.

Based on these works and following other empirical studies, the effect of transportation infrastructure on urban spatial structure can be examined by estimating a type of population density function[4]:

$$lnD_{it} = A_0 + \gamma_{inf} d_{inf,it} + A_1 X_{it} + \epsilon_{it} \quad (1)$$

where lnD_{it} is the natural logarithm of population density for census tract i in year t and $d_{inf,it}$ is the distance from the tract centroid to the transportation infrastructure. γ_{inf} is the transportation infrastructure gradient and measures whether population density increases or decreases with distance to this infrastructure. X denotes a vector of observed census tract characteristics such as location, geography, and history. ϵ is the error term.

To study whether transportation improvements affect changes in urban spatial structure, I estimate a first-difference specification based on Eq. (1):

[2] Non-monocentric/polycentric models allow the possibility of several main centers (Fujita and Ogawa, 1982; Anas and Kim, 1996; Lucas and Rossi-Hansberg, 2002). Since they adapt the AMM-type transportation infrastructure, the resulting spatial distribution of population only follows a decreasing density pattern from the CBD (the CBD gradient) and from the subcenters (subcenters' gradients).

[3] Baum-Snow (2007b) also extends Anas and Moses (1979) by allowing for different "technologies" to access the highways from the streets.

[4] This functional form is the linearized version of a negative exponential function derived from a quasilinear utility (Baum-Snow, 2007a). Monocentric density functions only include the distance to the CBD as explanatory variable (Clark, 1951; McDonald, 1989). Polycentric studies also consider the distance to the nearest subcenter (Anas et al., 1998).

Table 1
Summary statistics for main variables.

	Entire BMR		CBD		Subcenters		Non-central	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
1991–2006 Δln(population density)	0.034	0.471	−0.060	0.477	0.016	0.276	0.353	0.455
1991 ln(population density)	5.199	1.753	5.840	0.997	5.626	0.818	2.757	2.118
Distance to CBD (km)	11.653	11.078	5.708	3.352	17.515	11.182	25.342	12.079
Distance to the nearest subcenter (km)	6.638	4.567	6.934	3.669	1.987	1.365	9.935	5.571
Ruggedness index (m)	28.590	25.836	25.037	22.938	20.854	11.644	47.033	34.245
Distance to coast (km)	6.443	6.504	3.403	1.669	10.681	7.830	12.312	8.438
1991–2001 Changes in dist to hwy ramp (km)	−1.202	2.338	−0.957	1.062	−1.059	2.250	−2.116	4.326
Distance to the nearest railroad station (km)	0.849	1.359	0.399	0.323	0.915	0.697	2.228	2.474
Distance to the nearest roman road (km)	1.394	1.875	0.899	0.700	2.384	3.114	2.076	2.334
Distance to the nearest 19th century main road (km)	1.804	2.129	0.935	0.758	2.959	2.258	3.535	3.175
Distance to the nearest 19th century railroad (km)	1.479	1.625	0.917	0.758	1.986	1.201	2.816	2.706
Number of census tracts	3182		1992		568		622	

$$\Delta(lnD_{it}) = B_0 + \beta_{\gamma_{inf}}\Delta(d_{inf,it}) + B_1\Delta X_{it} + \eta_{it} \quad (2)$$

where $\Delta(lnD_{it}) = lnD_{it} - lnD_{it-1}$ measures population (density) growth. $\Delta(d_{inf,it}) = d_{inf,it} - d_{inf,it-1}$ are the changes in distance to the transportation infrastructure and measure its improvements. $\beta_{\gamma_{inf}}$ is a "growth" gradient and shows whether population (density) increases or decreases near transportation improvements. The abovementioned census tract characteristics are time-invariant and, as a result, drop out of this equation. Thus I include these characteristics in level and the initial population density as control variables to account for the possibility that initial conditions may determine population (density) growth and be correlated with transportation improvements.

Under the assumption that the random element of population (density) growth is uncorrelated with transportation improvements, I can estimate Eq. (2) by ordinary least squares (OLS). However, as Baum-Snow (2007a) points out, transportation improvements are expected to be endogenous to population (density) growth. That is, causation may either run from transportation improvements to population (density) growth or the converse may hold. Planners may want to serve areas with high predicted population growth or, alternatively, with poor prospects. In both cases, reverse causation would be at work.

To deal with this identification issue, I model transportation improvements explicitly in Eq. (3), a two-equation model, one to predict transportation improvements, the other to predict the effect of these improvements on population (density) growth:

$$\begin{aligned}\Delta(d_{inf,it}) &= C_0 + C_1\Delta X_{it} + C_2 Z_{it} + \mu_{it}\\ \Delta(lnD_{it}) &= B_0 + \beta_{\gamma_{inf}}\Delta(\widehat{d_{inf,it}}) + B_1\Delta X_{it} + \eta_{it}\end{aligned} \quad (3)$$

where $\Delta(\widehat{d_{inf,it}})$ is predicted changes in distance to the transportation infrastructure as estimated in the first-stage. Z_{it} are the exogenous instruments that have to predict transportation improvements and being otherwise uncorrelated with population (density) growth. That is, instruments have to satisfy the relevance, $\text{cov}(Z, \Delta(d_{inf,it})|X) \neq 0$, and the exogeneity, $\text{cov}(Z, \eta|X) = 0$, conditions.

3. Data

I center my analysis on the Barcelona Metropolitan Region which shows at least two relevant characteristics. First, the region is undergoing a process of population suburbanization that is affecting the central city, which loses inhabitants, and subcenters and, in particular, non-central suburban areas, which increase their population (Garcia-López, 2010). Second, the BMR transportation infrastructure is based on both a railroad network and a highway system. The former was constructed in the 19th century whereas the latter was constructed during the last quarter of the 20th century. Among other transportation improvements made between 1991 and 2001, the highway system was extended by 191 km and 104 new ramps were added. These changes were concurrent with the suburbanization process.

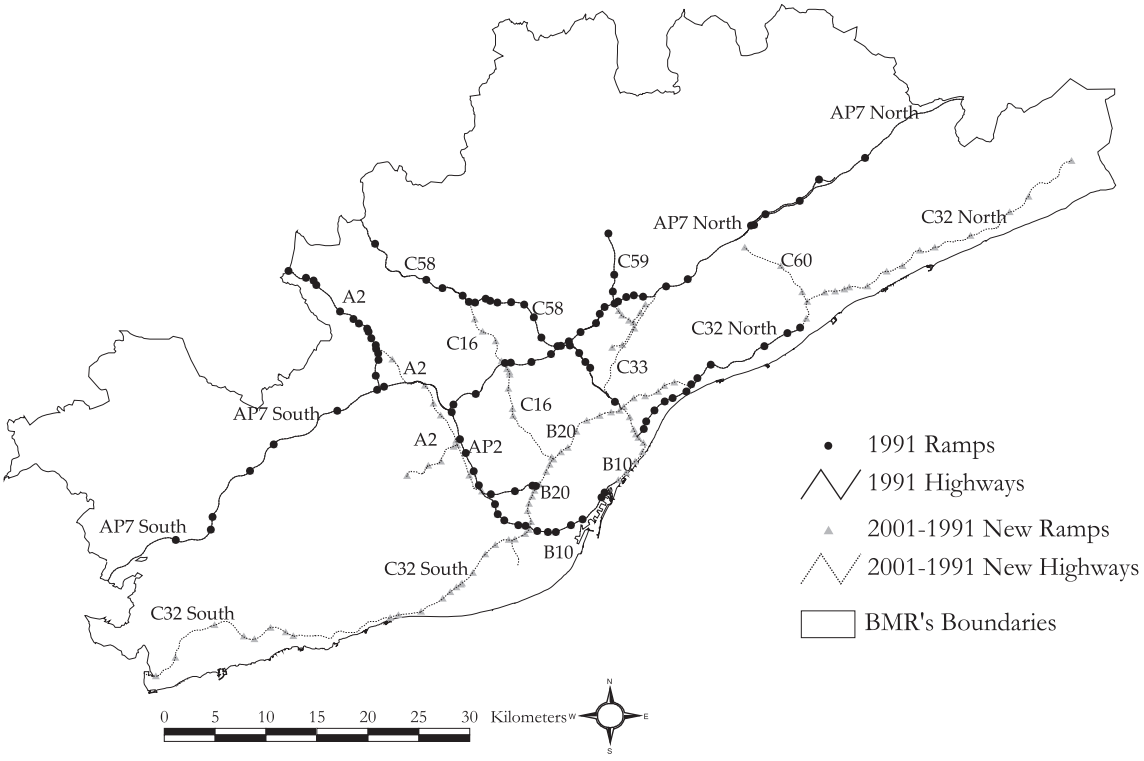
I use the census tract as my unit of observation which included 3182 tracts in the entire BMR sample and 1,992, 568 and 622 tracts in the CBD, subcenters and non-central suburban areas samples, respectively. Table 1 reports summary statistics for my main variables and Appendix A provides further details about their computation and their associated data.

I use population data from the 1991 Population Census, the earliest census with data at a tract level, and from the 2006 Municipal Register, both produced by the Instituto Nacional de Estadística (INE). I compute population densities for each tract by dividing its inhabitants by its total land area. I construct my dependent variable as the 1991–2006 changes in log population densities, $\Delta(lnD_{1991-2006}) = lnD_{2006} - lnD_{1991}$. I use tracts "designed" with the same boundaries in 1991 and 2006; thus total land area did not change during this time period and, consequently, my dependent variable can also be interpreted in terms of population growth. The related values in Table 1 show an slight increase of population (density) in the BMR between 1991 and 2006, but also indicate a process of suburbanization with population decreasing in the CBD and increasing in the subcenters and, in particular, in non-central suburban areas. To control for initial density conditions, I also use the natural logarithm of 1991 population density.

I control for other initial conditions by including a number of variables describing census tract location, physical geography, and history. Following the large literature on urban spatial structure, I use the census tract distances to the CBD and to the nearest subcenter to characterize its location with respect to main centers. Since physical geography may impact population (density) growth and old and new transportation infrastructure, I use an index of terrain ruggedness and distance to the coast. To control for history I also include the natural logarithm of municipal population every 20 years from 1910 to 1970.

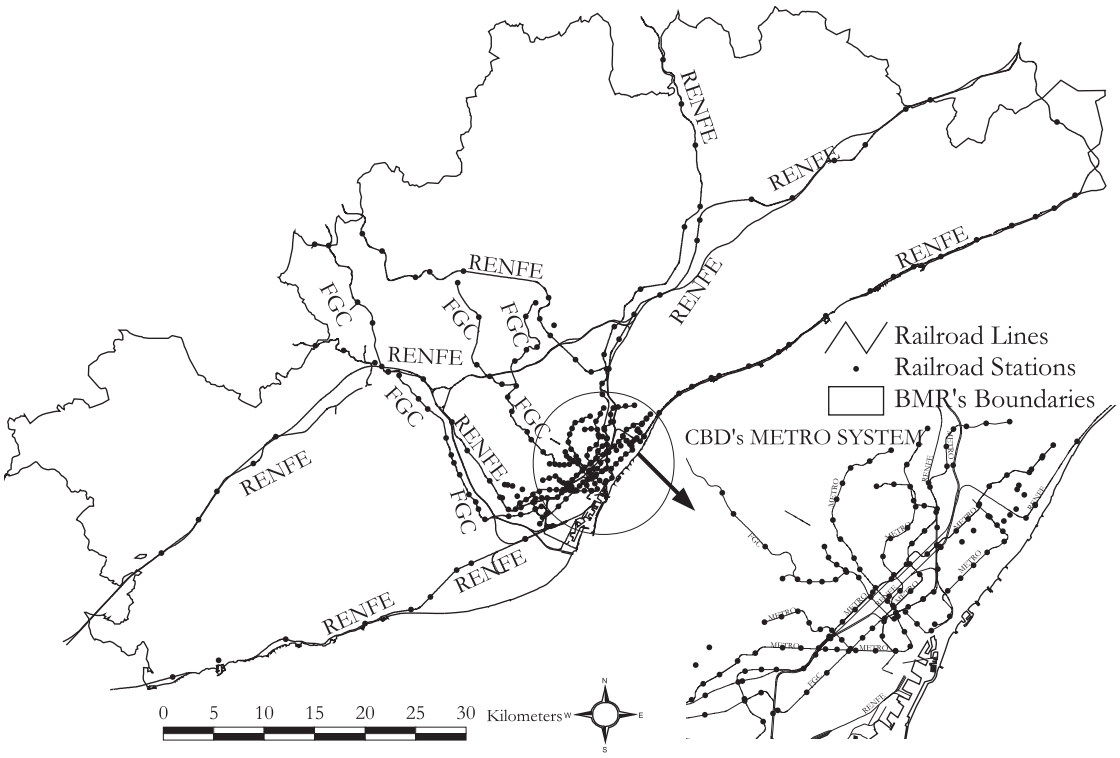
My main explanatory variables include one measure of transportation improvements related to the highway system and another related to the railroad network. Since the most recent transportation improvements are only related to the highway system between 1991 and 2001, I construct the highway variable as the 1991–2001 changes in distance to the nearest highway ramp ($\Delta(d_{hwy,1991-2001}) = d_{hwy,2001} - d_{hwy,1991}$) and the railroad variable as the distance to the nearest railroad station (d_{rail}).[5] To calculate these distances, I use vector digital maps related to the highway

[5] Although physically the railroad network did not change, its time schedules did and, as a result, train frequency was improved.



Source: Own elaboration.

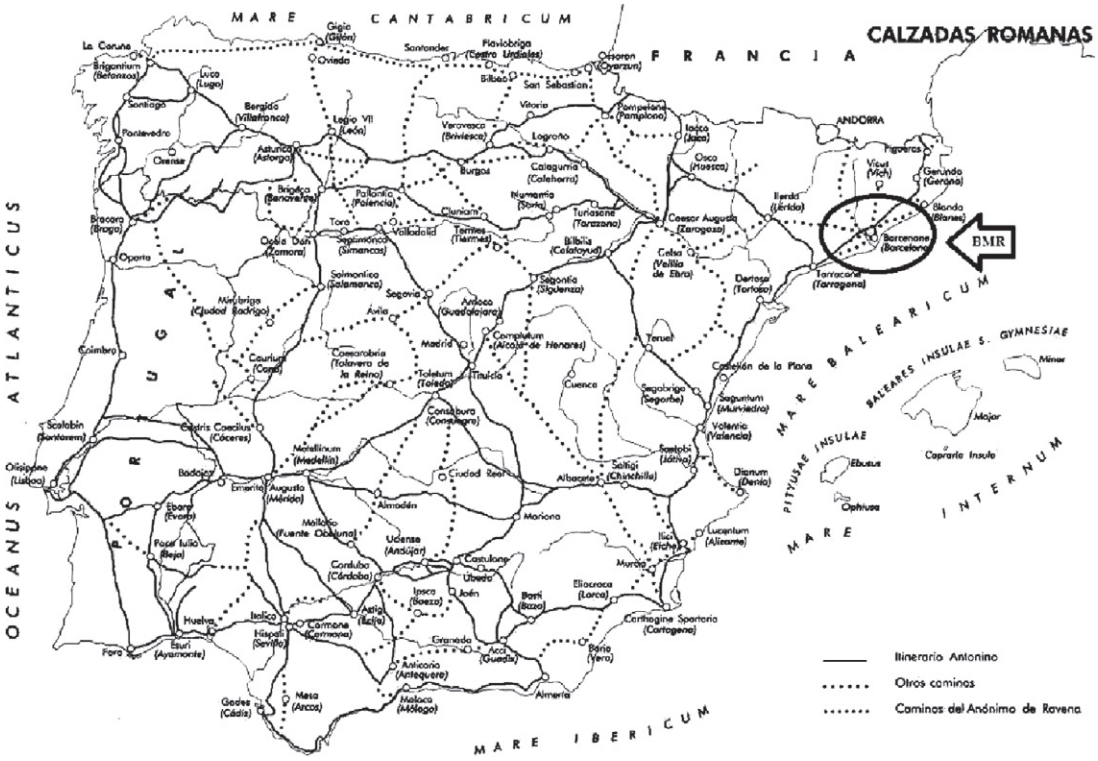
Fig. 1. The BMR highway system.



Source: Own elaboration.

Fig. 2. The BMR railroad network.

system and the railroad network in 2008 created by the Departament de Política Territorial i Obres Públiques (DPTOP). Based on these maps and other information collected from Instituto Geográfico Nacional (IGN) (IGN, 1991, 2008), I created the 1991 and 2001



Source: Atlas Nacional de España ©Instituto Geográfico Nacional de España (IGN, 2008).

Fig. 3. Roman roads in Spain.



Source: Atlas Nacional de España ©Instituto Geográfico Nacional de España (IGN, 2008).

Fig. 4. Spanish 19th century main roads.



Source: Atlas Nacional de España ©Instituto Geográfico Nacional de España (IGN, 2008).

Fig. 5. Spanish 19th century railroad network.

highway system and the 2006 railroad network maps. These are vector digital maps with polylines (highway and railroad segments) and points (ramps and stations) (Figs. 1 and 2). I compute straight-line distances (km) between each tract centroid and (1) the nearest 2001 highway ramp centroid; (2) the nearest 1991 highway ramp centroid; and (3) the nearest 2006 railroad station centroid.

The BMR highway system is based on 427 km of fast and high-capacity main roads (A2, AP2, AP7, C16, C32, C33, C58, C60, B10 and B20) where access is limited by 207 ramps located in 68 municipalities (Fig. 1). Although construction of the BMR highway system began in the late 1960s,[6] the movement to build a substantial highway system began with a national plan called "Plan General de Carreteras" in 1983 and with a state plan called "Pla de Carreteres de Catalunya" in 1985. The first plan focused on connecting the largest Spanish cities with the capital, Madrid, and by doing so relieving the traffic situations of the most congested corridors (Holl, 2011). The second plan centered on connecting metropolitan and non-metropolitan Catalan cities with Barcelona in order to improve their accessibility levels and to contain the spread of metropolitan growth.

As mentioned above, the most recent highway improvements are from the 1991 to 2001 period, when several highways were extended (e.g., the C32 North and A2 highways), new ramps were added (e.g., the AP7 North highway), and new highways were built (e.g., the C32 South, C33 and C60 highways). Furthermore, the first CBD highways were built crossing Barcelona municipality through its mountains (the B20 highway or Ronda de Dalt) and through its seafront (the B10 highway or Ronda Litoral). As a whole, 191 km of new highways and 104 new ramps were built between 1991 and 2001, and the distance to the nearest ramp reduced by 1.2 km for the entire BMR, by 1.0 km in central areas (CBD and subcenters) and by 2.1 km in non-central suburban areas.

The BMR railroad system is based on an important subway (METRO), and some railroad lines (FGC and RENFE) (Fig. 2). The subway covers the whole municipality of Barcelona and partially the CBD and is owned by a metropolitan public firm (Transports Metropolitans de Barcelona S.A.). The FGC railroad lines are owned by a state public firm (Ferrocarrils de la Generalitat de Catalunya S.A.) and provide services to Barcelona, to some CBD municipalities, and to some "counties" nearest to Barcelona. The RENFE railroad lines are part of the national railroad system and connect Barcelona with other BMR municipalities (i.e., metropolitan services) and the rest of the country (i.e., long-distance services). The first railroad line was inaugurated in 1848 and connected Barcelona and Mataró, one of its main subcenters.[7] The basic network was completed during the second half of the 19th century and most of the non-subway stations were built during that period. The first subway line was inaugurated in 1924 and most of its current stations were built before the 1992 Olympic Games (Miralles, 1997).[8]

Barcelona railroad network is mainly a passenger-oriented infrastructure with 1172 km of railroad lines and 282 stations in 76 municipalities. The central city, Barcelona, concentrates 128 railroad stations and there are 185 in the whole CBD (165 are subway stations). On average, the BMR census tracts are less than 1 km from the nearest railroad station. As expected, central census tracts are closer to these stations (400 m in the CBD and 900 m in the

[6] The first highway was the Maresme highway (the C32 North highway, formerly named the A-19 highway) and was opened in 1969. The northern stretch of the AP-7 highway was opened in 1970 and the southern stretch in 1972. The AP2 highway between Barcelona and Martorell was opened in 1972 (Miralles, 1997).

[7] In fact, this was also the first Spanish railroad line. Ulterior BMR railroad lines connected Barcelona with other subcenters such as Granollers (1854), Sabadell (1855), or Martorell (1861).

[8] From 1992 to 2001 only 5 new stations were inaugurated. The new subway lines L9 and L10 are the most recent projects to improve the subway system adding 49 km and 52 new stations. They are under construction since 2006.

Table 2
Changes in population location patterns and improvements in highways and railroads, OLS – unconditional vs. conditional effects.

Dependent variable:	1991–2006 Δln(population density)							
Area:	Entire BMR		CBD		Subcenters		Non-central	
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
ΔDistance to the nearest H. ramp	−0.0203[a]	−0.0076[a]	0.0137[c]	0.0181[b]	−0.0195[a]	−0.0053	−0.0138[a]	−0.0098[a]
	(0.0033)	(0.0029)	(0.0080)	(0.0079)	(0.0067)	(0.0047)	(0.0040)	(0.0036)
Distance to the nearest R. station	0.0865[a]	−0.0177[a]	0.0450	−0.0797[c]	0.0439[c]	−0.0166	0.0383[a]	−0.0246[a]
	(0.0075)	(0.0068)	(0.0431)	(0.0422)	(0.0243)	(0.0175)	(0.0073)	(0.0082)
ln(1991 population density)		−0.1282[a]		−0.0983[a]		−0.2274[a]		−0.1300[a]
		(0.0064)		(0.0121)		(0.0163)		(0.0105)
Adjusted R^2	0.07	0.21	0.00	0.04	0.04	0.45	0.05	0.30

Notes: All regressions include a constant term. 3182, 1992, 568 and 622 observations for BMR, CBD, subcenters and non-central areas regressions, respectively. Robust standard errors in parentheses.
[a] Significant at the 1 percent level.
[b] Significant at the 5 percent level.
[c] Significant at the 10 percent level.

Table 3
Changes in population location patterns and improvements in highways and railroads, OLS – extended conditional effects.

Dependent variable:	1991–2006 Δln(population density)								
Area:	CBD			Subcenters			Non-central areas		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
ΔDistance to ramp	0.0179[b]	0.0105	−0.0040	−0.0083	−0.0033	−0.0026	−0.0179[a]	−0.0202[a]	−0.0196[a]
	(0.0073)	(0.0096)	(0.0095)	(0.0061)	(0.0061)	(0.0063)	(0.0045)	(0.0051)	(0.0054)
Distance to station	−0.0744	−0.0581	−0.0740	−0.0058	−0.0125	−0.0138	−0.0182[b]	−0.0171[b]	−0.0169[b]
	(0.0567)	(0.0531)	(0.0552)	(0.0212)	(0.0239)	(0.0255)	(0.0082)	(0.0083)	(0.0084)
ln(1991 pop density)	−0.0989[a]	−0.0991[a]	−0.0972[a]	−0.2392[a]	−0.2462[a]	−0.2449[a]	−0.1383[a]	−0.1433[a]	−0.1451[a]
	(0.0120)	(0.0119)	(0.0121)	(0.0212)	(0.0222)	(0.0223)	(0.0111)	(0.0142)	(0.0148)
Distances to CBD	Y	Y	Y	Y	Y	Y	Y	Y	Y
Geography	N	Y	Y	N	Y	Y	N	Y	Y
ln(past populations)	N	N	Y	N	N	Y	N	N	Y
Adjusted R^2	0.03	0.03	0.04	0.45	0.46	0.46	0.31	0.31	0.31

Notes: All regressions include a constant term. 1992, 568 and 622 observations for CBD, Subcenters and Non-Central Areas regressions, respectively. Robust standard errors in parentheses. [c]Significant at the 10 percent level.
[a] Significant at the 1 percent level.
[b] Significant at the 5 percent level.

subcenters) than census tracts in non-central suburban areas (2.2 km) (Table 1).

My instruments are based on old transportation infrastructures in Spain: Roman roads (Fig. 3), main roads (Fig. 4), and railroad lines (Fig. 5) constructed before the end of the 19th century. Duranton and Turner (2011, forthcoming) also use historic instruments such as the 19th century railroads and the 19th century major expedition routes. The use of Roman roads as an instrumental variable is new to the literature. All three instruments are straight-line distances in km from each tract centroid to the nearest segment of old transportation infrastructure. To compute distance I use vector digital maps based on de Soto (2010) (Roman roads) and Carreras and de Soto (2010) (19th century main roads and railroad lines), and maps and other information collected from Instituto Geográfico Nacional (IGN, 1991, 2008).

In order to use these instruments, they have to be exogenous and relevant. *A priori*, all three instruments are exogenous because of the length of time since these old infrastructures were built and the significant historical changes in the society and the economy (Duranton and Turner, forthcoming). In this sense, BMR suburbanization process is a relatively new phenomenon that emerged in the mid-1980s (Muñiz et al., 2003). It seems obvious that these old transportation networks were not built to anticipate either current density patterns or the current suburbanization process. On the contrary, the main reasons for building transportation networks were related to military, administrative, political and communication concerns, or stem from adaptation to the "new" transportation modes. Appendix B provides further details. Because these old infrastructures were not placed randomly and some of the factors that influenced their location may have also influenced modern transportation improvements, instrument exogeneity hinges on controlling for physical geography and historical population.

Conceptually, the relevance of these instruments rest on the fact that new transportation infrastructures are not built in isolation of previous and older infrastructures. In general, new infrastructures are easier and cheaper to build close to old infrastructures (Duranton and Turner, forthcoming). However, it is also possible that new infrastructures are far from the old networks because of differences in the main reason that motivated their construction (economic vs. political decisions) or due to the competition between transportation modes (road vs. railroads). For the case of Spain, Holl (2011) and Bel (2011) point out the critical role of these historical networks for the placement of the new transportation infrastructure.

4. Results

4.1. OLS results

I first estimate Eq. (2) by OLS. Although coefficient estimates are not valid due to the endogeneity problem, I use OLS results for descriptive purposes.

Table 2 presents OLS results describing the effects of the 1991–2001 changes in distance to the nearest highway ramp and the

Table 4
First-stage and reduced-form results, OLS.

Dependent variable	First-stage estimates						Reduced-form estimates		
	1991–2001 ΔDist to ramp			Dist to station			1991–2006 Δln(pop dens)		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Panel A. CBD									
Dist to roman road	−0.4592[a]	−0.2922[a]	−0.3625[a]	0.1496[a]	0.1456[a]	0.1414[a]	−0.0404[a]	−0.0300[b]	−0.0329[b]
	(0.0394)	(0.0354)	(0.0323)	(0.0181)	(0.0211)	(0.0209)	(0.0143)	(0.0147)	(0.0151)
Dist to 19th century road	0.1284[a]	0.1563[a]	0.1645[a]	−0.0732[a]	−0.0510[a]	−0.0644[a]	−0.0101	−0.0157	−0.0191
	(0.0417)	(0.0446)	(0.0361)	(0.0154)	(0.0156)	(0.0148)	(0.0171)	(0.0175)	(0.0200)
Dist to 19th century rail.	−0.9028[a]	−0.6017[a]	−0.7060[a]	−0.0140	−0.0125	0.0016	−0.0038	0.0120	0.0099
	(0.0475)	(0.0436)	(0.0381)	(0.0135)	(0.0184)	(0.0187)	(0.0134)	(0.0140)	(0.0133)
ln(1991 pop dens)	Y	Y	Y	Y	Y	Y	Y	Y	Y
Distances	Y	Y	Y	Y	Y	Y	Y	Y	Y
Geography	N	Y	Y	N	Y	Y	N	Y	Y
ln(past pop)	N	N	Y	N	N	Y	N	N	Y
Adjusted R^2	0.35	0.48	0.62	0.38	0.43	0.48	0.03	0.04	0.04
Panel B. Subcenters									
Dist to roman road	−0.4238[a]	−0.4755[a]	−0.4974[a]	−0.0289[a]	−0.0150[c]	−0.0093	0.0027	0.0040	0.0039
	(0.0461)	(0.0152)	(0.0161)	(0.0106)	(0.0090)	(0.0096)	(0.0035)	(0.0035)	(0.0036)
Dist to 19th century road	−0.5095[a]	−0.1042[a]	−0.1810[a]	0.1045[a]	0.1088[a]	0.0885[a]	−0.0141[a]	−0.0104[c]	−0.0102[d]
	(0.0412)	(0.0177)	(0.0214)	(0.0153)	(0.0156)	(0.0164)	(0.0051)	(0.0060)	(0.0068)
Dist to 19th century rail.	−0.5695[a]	−0.2392[a]	−0.1183[a]	0.1053[a]	0.1059[a]	0.1050[a]	−0.0187[b]	−0.0205[b]	−0.0293[a]
	(0.0814)	(0.0350)	(0.0436)	(0.0276)	(0.0244)	(0.0276)	(0.0094)	(0.0093)	(0.0110)
Adjusted R^2	0.44	0.84	0.86	0.37	0.47	0.48	0.47	0.47	0.47
Panel C. Non-central areas									
Dist to roman road	−0.4930[a]	−0.4470[a]	−0.4168[a]	0.2266[a]	0.1872[a]	0.1916[a]	−0.0012	0.0007	0.0008
	(0.0604)	(0.0428)	(0.0489)	(0.0406)	(0.0379)	(0.0388)	(0.0080)	(0.0081)	(0.0083)
Dist to 19th century road	0.2781[a]	0.1392[a]	0.0954[b]	0.1723[a]	0.2269[a]	0.2211[a]	−0.0236[a]	−0.0244[a]	−0.0224[a]
	(0.0409)	(0.0398)	(0.0411)	(0.0340)	(0.0328)	(0.0331)	(0.0055)	(0.0061)	(0.0062)
Dist to 19th century rail.	0.4984[a]	0.0789[c]	0.0510	0.3091[a]	0.3095[a]	0.3029[a]	0.0021	0.0129[c]	0.0134[c]
	(0.0475)	(0.0474)	(0.0502)	(0.0428)	(0.0512)	(0.0509)	(0.0060)	(0.0078)	(0.0079)
Adjusted R^2	0.47	0.57	0.61	0.56	0.60	0.61	0.30	0.31	0.31

Notes: All regressions include a constant term. 1992, 568 and 622 observations for regressions in Panels A, B and C, respectively. Robust standard errors in parentheses.
[a] Significant at the 1 percent level.
[b] Significant at the 5 percent level.
[c] Significant at the 10 percent level.
[d] Significant at the 20 percent level.

distance to the nearest railroad station on the 1991–2006 changes in log population densities. Odd columns 1, 3, 5, and 7 show unconditional results, that is, without any control variable. Coefficients on highways indicate that tracts with higher reductions in their distances to the highway system grow more quickly in the entire BMR (column 1) and, in particular, in suburban areas (columns 5 and 7). Simultaneously, these improvements have a negative impact on growth inside the CBD. Both results are consistent with the finding of Baum-Snow (2007a) that highway improvements cause suburbanization.

Except for the CBD, coefficients on railroads show that tracts located further away from stations grow more quickly. Since the railroad system is older and its stations are located in the most developed and dense tracts of each area, these results also show that when a new transportation infrastructure is built, development occurs first nearby. Nearby tracts then become fully developed and subsequent development takes place further away.

Even columns in Table 2 include initial population density as a control and thus present conditional results. Since coefficients on highway improvements vary only slightly and become insignificant for subcenters, the above interpretations apply: while highway improvements foster population growth in the entire BMR and particularly in non-central suburban areas, improvements have a negative impact on growth of CBD tracts.

Coefficients for railroad stations dramatically change and become negative. This result indicates that when conditioned on density, proximity to stations has a positive effect on tract growth in general and on non-central suburban growth in particular. This conditional effect complements the first finding of a negative unconditional effect: on average, each additional kilometer closer to the nearest station reduces population growth, but conditioning out the fact that nearly everything around the station is dense and developed and that high density tracts grow more slowly, proximity to station still has a positive effect on nearby growth.

Since BMR results represent estimates of the average, I now turn my attention to three extended area results.[9] In Table 3, I gradually include controls for census tract location (columns 1, 4 and 5), physical geography (columns 2, 5 and 8), and history (columns 3, 6 and 9). Adding these controls only changes the significance of CBD coefficients on both transportation variables, which become insignificant. Thus these extended results confirm the previous findings: railroad proximity and highway improvements have a positive effect on population growth only in non-central areas. Since the Barcelona Metropolitan Region is undergoing a suburbanization process, these results also indicate that transportation improvements cause suburbanization and affect its spatial pattern.

4.2. IV results

4.2.1. First-stage and reduced-form results

To econometrically test the relevance of my candidates as instruments, I implement the strategy of Murray (2006a,b).

[9] BMR results are similar to non-central results and are in Appendix Table C.1.

Table 5
Changes in population location patterns and improvements in highways and railroads, IV.

Dependent variable	1991–2006 Δln(population density)				
	[1]	[2]	[3]	[4]	[5]
Panel A. CBD					
Instruments: Distances to the nearest roman road and to the nearest 19th century main road					
1991–2001 ΔDist to ramp	0.0768	−0.0685	−0.1016	−0.0654	−0.3290
	(0.0625)	(0.0815)	(0.1005)	(0.0568)	(0.2421)
Distance to Station	0.0064	−0.4192[c]	−0.4937[c]	−0.2649[b]	−0.8067[c]
	(0.1474)	(0.2210)	(0.2781)	(0.1110)	(0.4721)
ln(1991 population density)		−0.1388[a]	−0.1437[a]	−0.1262[a]	−0.1394[a]
		(0.0261)	(0.0299)	(0.0173)	(0.0299)
Distances	N	N	Y	Y	Y
Geography	N	N	N	Y	Y
ln(past populations)	N	N	N	N	Y
First-stage statistic	16.59	8.70	5.08	24.29	2.56
First-stage statistic ramp	61.81	38.57	28.07	56.37	8.57
First-stage statistic station	37.53	31.49	19.47	50.16	8.40
Panel B. Subcenters					
Instruments: Distances to the nearest 19th century main road and to the nearest 19th century railroad					
1991–2001 ΔDist to ramp	−0.2106[a]	−0.0904	0.0041	−0.0297	−0.0343
	(0.0556)	(0.0734)	(0.0212)	(0.0289)	(0.0284)
Distance to station	0.2412[a]	0.0007	−0.1550[a]	−0.1732[b]	−0.2503[b]
	(0.0745)	(0.1128)	(0.0535)	(0.0689)	(0.0986)
ln(1991 population density)		−0.1778[b]	−0.2323[a]	−0.2235[a]	−0.2171[a]
		(0.0716)	(0.0250)	(0.0313)	(0.0327)
First-stage statistic	12.72	1.59	15.62	15.85	9.47
First-stage statistic ramp	33.77	5.09	38.18	52.18	35.95
First-stage statistic station	115.36	16.59	42.35	38.08	22.55
Panel C. Non-central areas					
Instruments: Distances to the nearest roman road and to the nearest 19th century main road					
1991–2001 ΔDist to ramp	−0.0868[b]	−0.0503[c]	−0.0240[c]	−0.0355[c]	−0.0392[c]
	(0.0432)	(0.0318)	(0.0123)	(0.0194)	(0.0224)
Distance to station	−0.0045	−0.0843[a]	−0.0547[a]	−0.0746[a]	−0.0740[a]
	(0.0166)	(0.0233)	(0.0184)	(0.0213)	(0.0226)
ln(1991 population density)		−0.1682[a]	−0.1629[a]	−0.1652[a]	−0.1614[a]
		(0.0183)	(0.0148)	(0.0170)	(0.0169)
First-stage statistic	4.22	4.71	75.45	35.73	21.39
First-stage statistic ramp	9.56	10.82	133.74	106.95	63.36
First-stage statistic station	78.91	55.76	72.07	70.15	63.60

Notes: All regressions include a constant term. 1992, 568 and 622 observations for regressions in Panels A, B and C, respectively. Robust standard errors in parentheses. Endogenous variables are instrumented simultaneously.
[a] Significant at the 1 percent level.
[b] Significant at the 5 percent level.
[c] Significant at the 10 percent level.

Specifically, I estimate reduced-form equations that predict transportation improvements (first-stage estimates) and population (density) growth (reduced-form estimates) as a function of each instrument candidate. In general, I choose as an instrument the candidate with coefficients significantly different from zero and with signs that support intuition: positive or negative first-stage results indicates that transportation improvements are close to or far from the instrument, while negative reduced-form results indicates that proximity to the instrument fosters population growth.

Columns 1–3 of Table 4 presents first-stage results for the 1991–2001 changes in distance to the nearest highway ramp, and columns 4–6 show results for the distance to the nearest railroad station. Columns 7–9 show reduced-form results. Each set of regressions includes the three extended conditional specifications. Panels A, B, and C report results for CBD, subcenters, and non-central suburban areas, respectively.

The selection of instruments is more straight-forward for regressions of subcenters. Following the abovementioned strategy, I choose the distances to the nearest 19th century main road and to the nearest 19th century railroad as instruments. First-stage results indicate that highways are improved far from these 19th century infrastructures, whereas modern stations are close to them. Reduced-form estimates indicate that proximity to these old networks has a positive effect on population (density) growth.

As for CBD regressions the selection is not so clear because only one candidate fulfills both criteria: the distance to the nearest Roman road. Since I have two endogenous variables, I choose the distance to the nearest 19th century main road as my second instrument because all first-stage coefficients are significant for this variable. For selected instruments, first-stage results indicate that highway improvements are far from Roman roads and close to 19th century main roads, whereas railroad stations are near Roman roads and far from 19th century main roads. Reduced-form estimates indicate that proximity to Roman roads has a positive effect on population (density) growth.

Similarly, I select the distance to the nearest 19th century main road as instrument for non-central suburban regressions. Although not satisfying the reduced-form criterion, I also select the distance to the nearest Roman road. Their first-stage results indicate that highways are improved far from Roman roads and near 19th century main roads, whereas modern stations are close to both old infrastructures. Reduced-form estimates show that proximity to the 19th century main roads fosters population growth. I do not choose distance to the 19th century railroad because its reduced-form

Table 6
Changes in population location patterns and 1991 highways and railroads, OLS and IV: non-central suburban areas.

Dependent variable	1991–2006 Δln(population density)					
	OLS			IV		
	[1]	[2]	[3]	[4]	[5]	[6]
Instr.: Roman and 19th century roads						
1991 Distance to ramp	0.0122[a]	0.0118[b]	0.0108[b]	0.0321[c]	0.0529[c]	0.0704[c]
	(0.0045)	(0.0046)	(0.0049)	(0.0169)	(0.0314)	(0.0474)
Distance to station	−0.0289[a]	−0.0247[a]	−0.0233[a]	−0.0860[a]	−0.1310[a]	−0.1553[b]
	(0.0080)	(0.0085)	(0.0087)	(0.0216)	(0.0454)	(0.0689)
ln(1991 population density)	−0.1357[a]	−0.1421[a]	−0.1448[a]	−0.1680[a]	−0.1784[a]	−0.1753[a]
	(0.0111)	(0.0143)	(0.0149)	(0.0161)	(0.0227)	(0.0235)
Distances	Y	Y	Y	Y	Y	Y
Geography	N	Y	Y	N	Y	Y
ln(past populations)	N	N	Y	N	N	Y
Adjusted R^2	0.30	0.30	0.30			
First-stage statistic				31.80	10.52	4.09
First-stage statistic ramp				72.91	34.54	15.45
First-stage statistic station				51.95	24.21	12.73

Notes: All regressions include a constant term. 622 Observations for each regression. Robust standard errors in parentheses. Endogenous variables are instrumented simultaneously.
[a] Significant at the 1 percent level.
[b] Significant at the 5 percent level.
[c] Significant at the 10 percent level.

results do not support intuition: its positive coefficients indicate that proximity to this infrastructure has a negative effect on growth.

4.2.2. TSLS results

Table 5 presents two stage least squares (TSLS) estimates for Eq. (3). Column 1 includes only transportation variables. Column 2 adds initial density as control, column 3 adds location variables, column 4 adds physical geography variables, and columns 5 adds municipal populations every 20 years from 1910 to 1970. Panels A, B, and C report results for CBD, suburban subcenters, and non-central suburban areas, respectively, using their selected instruments.

Table 5 also reports first-stage F-statistics for my selected instruments.[10] Since I have two endogenous variables, I present a global F-statistic and two individual F-statistics, one for each endogenous variable. For my unconditional specification in column 1 and my preferred conditional specifications in columns 4 and 5, all individual and most global F-statistics pass the size and relative bias tests proposed by Stock and Yogo (2005) in the context of TSLS estimations.[11]

In general, TSLS results and their OLS counterparts (in Tables 2 and 3) mostly coincide in coefficient interpretation, but differ in magnitude and significance of some variables. If I restrict the attention to unconditional results (column 1), TSLS coefficients on highways confirm the positive effect of their improvements on growth and are, on average, 10.8 and 6.3 larger (in absolute values) for subcenters and non-central regressions, respectively. This variable becomes insignificant for CBD regressions. TSLS results also confirm the negative impact of proximity to railroad stations on growth, but only for subcenters.

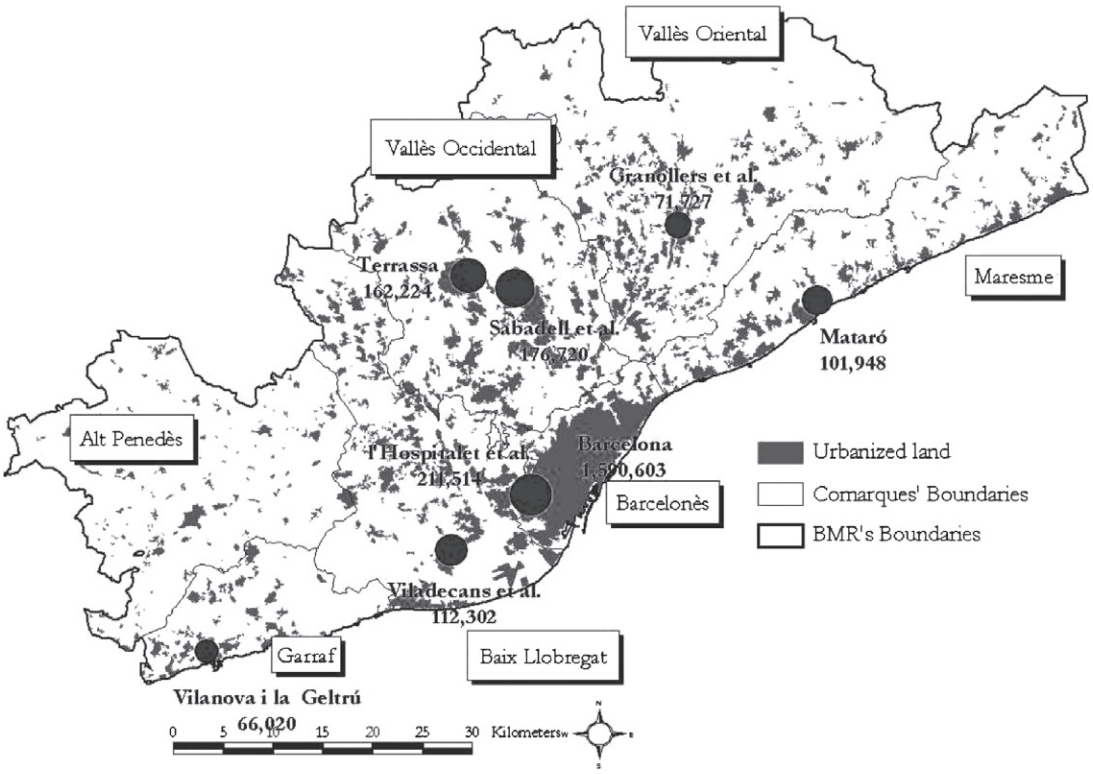
For conditional results (columns 2–5), TSLS coefficients on stations dramatically increase by an average factor of 27.4 and 14.8 for CBD and subcenters regressions, respectively.[12] Most importantly, these coefficients are now significant for central areas regressions, indicating that proximity to stations have a positive effect on growth in such areas. TSLS coefficients on railroads and highway improvements remain significant and confirm their positive effect in non-central areas. On average, they are 3.7 and 2.5 larger than their corresponding OLS estimates.

In summary, TSLS results indicate that transportation affects urban spatial structure. In particular, transportation improvements cause suburbanization and influence the spatial pattern of this process by attracting population to non-central suburban tracts that have had improved access to the highway system and railroad stations. Inside the CBD, transportation also affects the spatial distribution of residences by attracting population near railroad stations.

Two interesting OLS findings associated the effect of transportation infrastructures to the length of time since they were built and the degree of land development. TSLS results in turn verify these two findings. In particular, unconditional results indicate that population increases in undeveloped tracts close to new transportation infrastructures (i.e., highways) and far from old transportation infrastructures (i.e., railroads) where nearby tracts are fully developed. Conditioned on land development (density), TSLS results also suggest that population nevertheless increases near old infrastructures (railroads).

In Table 6 I explore in more detail the second finding (i.e., the role of old infrastructures) by estimating the conditional effect of the 1991 distance to the nearest highway ramp and the distance to the nearest railroad station on the 1991–2006 changes in log population densities. For the non-central area, in columns 1–3- I present OLS regressions, while in columns 4–6 I report TSLS regressions. These regressions are similar to their OLS and TSLS counterparts in Table 2 (columns 7–9) and Table 5 (columns 3–5 Panel C), respectively.[13] Results suggest that the second finding still holds for railroads, but not for highways: proximity to 1991 highway ramps has a negative effect on non-central growth. These opposing effects of transportation infrastructures may be due to differences in types

[10] F-statistics are computed from first-stage regressions that only include my selected instruments. Results approximate those in Table 4 and are available upon request.
[11] Specifications 5 in Panel A and 1 in Panel C do not pass weak instrument tests. Limited information matixum likelihood (LIML) estimates provide identical results.
[12] Despite those increments, TSLS-OLS differences are not economically significant. As shown in Table 1, population decreases (−6.0%) and slightly increases (1.6%) in CBD and subcenters, respectively.
[13] Table 6 also reports first-stage F-statistics for my selected instruments. Specification 6 does not pass weak instrument tests. Its LIML estimates provide identical results. First-stage and reduced-form results for selecting instruments are in Appendix Table C.2.



Source: Own elaboration.

Fig. A.1. Polycentricity in Barcelona: CBD and subcenters, 2006.

of residential development: while tracts around railroad stations show high dwelling densities and can accommodate more growth, residential development in tracts near highway ramps is based on low dwelling densities.

5. Conclusions

This paper investigates the effect of transportation improvements on changes in population location patterns in metropolitan Barcelona between 1991 and 2006. The analyses presented here make a novel contribution to the literature by using data on a much finer geographical scale (i.e., census tracts), by studying three city areas (i.e., the CBD, suburban subcenters, and non-central suburban areas), and by considering two types of transportation infrastructure (i.e., the highway system through its ramps and the railroad network through its stations).

Because of the endogeneity of infrastructure provision, I rely on IV techniques in order to obtain unbiased estimates. Another novel feature of this study is the use of three historic instruments at the intrametropolitan level: the Roman roads, the 19th century main roads, and the 19th century railroads. To my knowledge, the use of Roman roads as instrumental variable is new to the literature.

Main results verify and extend the finding of Baum-Snow (2007a) that transportation cause suburbanization: both highway and railroad improvements foster population growth in suburban areas, whereas the transit system also affects the location of population inside the CBD.

Results also indicate that the effect of transportation infrastructures depends on the length of time since construction, the degree of land development and the type of residential development. Specifically, population increases in undeveloped tracts close to new transportation infrastructures (1991–2001 highways) and far from old transportation infrastructures (railroads and 1991 highways) where nearby tracts are fully developed. If residential development of those nearby tracts is based on high dwelling densities, old infrastructure (railroads) still has a positive effect on nearby growth. If residential development is based on low dwelling densities, population increases far from the old infrastructure (1991 highways).

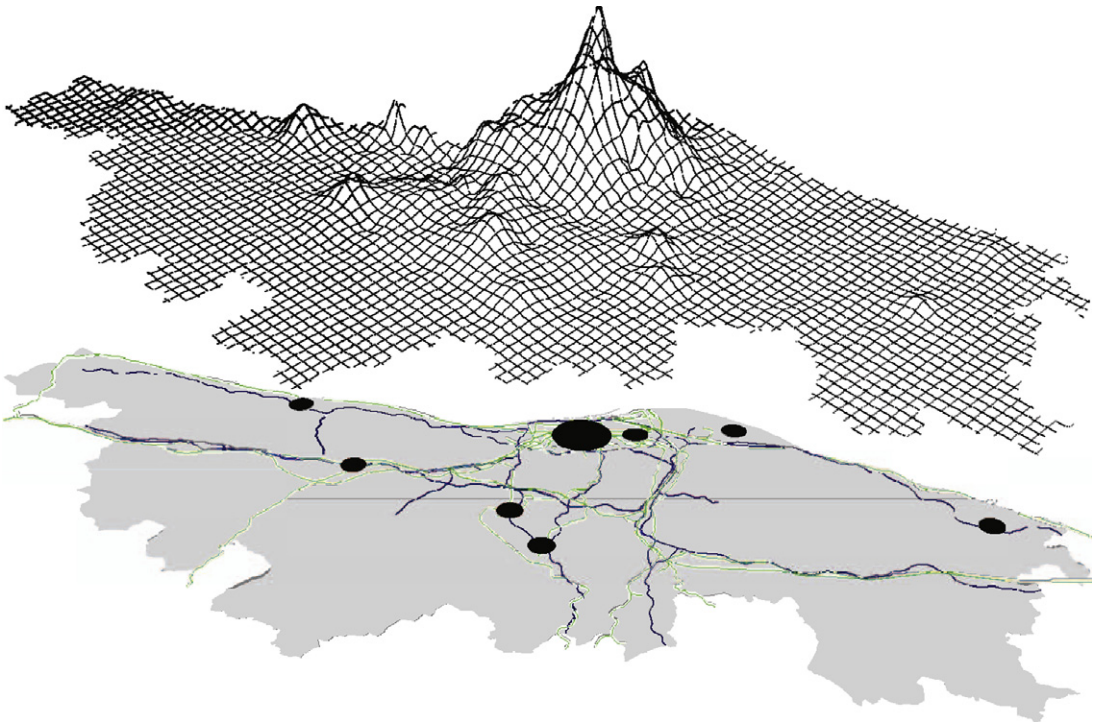
I leave for future research two outstanding issues related to this paper. Since this paper is centered on changes in intrametropolitan location patterns, the first issue concerns the effect of transportation improvements on residential land consumption patterns. Moreover, Barcelona is also undergoing a process of employment decentralization that might affect the process of population suburbanization. Thus the second issue is to simultaneously analyze the effect of transportation improvements on both processes.

Appendix A. Barcelona data

The Barcelona Metropolitan Region is one of the most important European metropolises. The city was delimited in 1966 by the Pla Director de l'Àrea Metropolitana, defined by law in 1987 by Lleis d'Ordenació Territorial, and it is currently used as a functional region for Planning purposes by the Pla Territorial General de Catalunya (PTGC).[14] Comprised of 164 municipalities grouped in 7 "comarques" or counties, the BMR covers an area of 324,000 ha in a radius of approximately 55 km. Because of its steep topography, only 67,999 ha were urbanized in 2002.[15]

[14] The PTGC is the basic planning tool in Catalonia and where 6 other functional regions are represented. For more information, see http://ca.wikipedia.org/wiki/Pla_territorial_general_de_Catalunya.

[15] See Muñiz et al. (2008), Garcia-López and Muñiz (2010), Muñiz and Garcia-López (2010) or Garcia-López and Muñiz (forthcoming), among others, for further details.



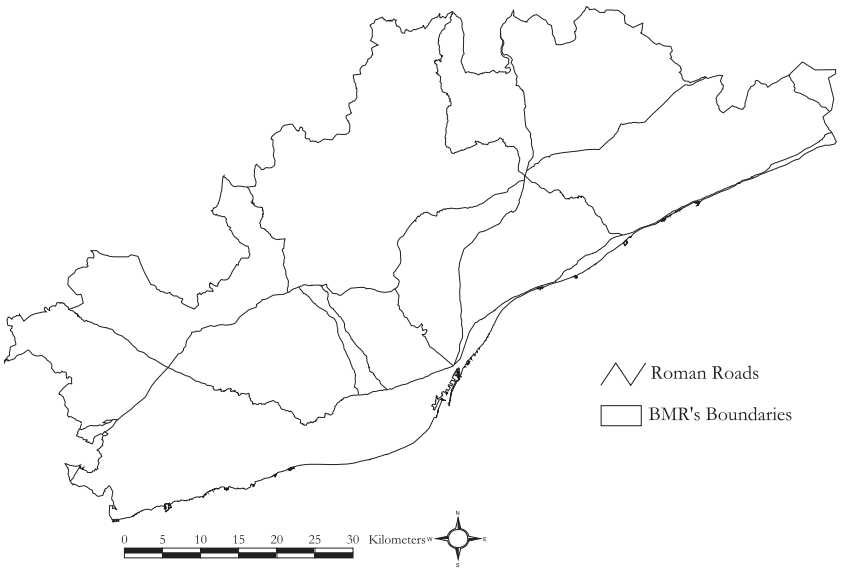
Source: Own elaboration.

Fig. A.2. Polycentricity in Barcelona: density and infrastructure, 2006.



Source: Atlas Nacional de España ©Instituto Geográfico Nacional de España (IGN, 2008).

Fig. B.1. The Spanish 18th century postal routes.



Source: Own elaboration based on de Soto (2010).

Fig. B.2. Roman roads of the BMR.

The BMR has been characterized as polycentric by Garcia-López (2010). Defining the Barcelona municipality as the main center, Garcia-López (2010) identifies seven main subcenters in 2005 (Fig. A.1). Using census tract data, main centers are identified applying the non-parametric methodology based on locally weighted regression (LWR) and developed by McMillen (2003). The idea is to estimate a density function that represents a monocentric spatial structure. Using it as a benchmark, the groups of significant deviations from the monocentric configuration (i.e., real densities greater than estimated densities) that surpass a cut-off are defined as centers.

Fig. A.2 is upside-down and shows the 2006 estimated three-dimensional density pattern overlapping BMR transportation infrastructure. Using GIS software, the 3D profile is estimated by LWR using an inverted distance weight function and a 1% "neighborhood".[16]

Since the number of Barcelona census tracts increased between 1991 and 2006 from 3569 to 3864, I merge tracts using related shapefiles to obtain a total of 3182 census tracts with identical census boundaries.

To compute location variables, I consider the CBD and 7 subcenters identified by Garcia-López (2010). I use GIS software to obtain tract centroid coordinates and calculate distances in straight line (km) from each centroid to the CBD central tract centroid and to the nearest subcenter central tract centroid.

To measure physical geography, I compute the terrain ruggedness index developed by Riley et al. (1999) using the Spanish 200-m digital elevation model (`http://www.ign.es/ign/layoutIn/modeloDigitalTerreno.do`). This variable is the average terrain ruggedness index for each census tract. I also compute distance in kilometers from each tract centroid to the coast. I do not include tract altitude as an explanatory variable because it is highly correlated with the terrain ruggedness index (60%) and distance to the coast (80%).

History variables are past populations computed at a municipal level because census tract data were not available until 1991. These data come from 1910, 1930, 1950, and 1970 Population Censuses produced by the Instituto Nacional de Estadìstica (INE).

[16] See McMillen (2001), Redfearn (2009) and McMillen and Redfearn (2010) for details on the LWR procedure.

Appendix B. Old transportation infrastructures as instruments

My three instruments are based on old transportation infrastructures in Spain: the Roman roads, the 19th century main roads, and the 19th century railroad lines.

Like many European cities that were either founded by Romans or were later subsumed by the Roman Empire, the very beginning

Table C.1
Changes in population location patterns and improvements in highways and railroads, OLS – extended conditional effects: BMR.

Dependent variable	1991–2006 Δln(population density)		
Area	Entire BMR		
	[1]	[2]	[3]
1991–2001 ΔDistance to the nearest highway ramp	−0.0054[c]	−0.0090[b]	−0.0096[a]
	(0.0030)	(0.0036)	(0.0037)
Distance to the nearest railroad station	−0.0220[a]	−0.0174[b]	−0.0164[b]
	(0.0071)	(0.0073)	(0.0073)
ln(1991 population density)	−0.1255[a]	−0.1293[a]	−0.1271[a]
	(0.0070)	(0.0075)	(0.0080)
Distance to CBD	0.0023[a]	0.0015[c]	0.0004
	(0.0008)	(0.0009)	(0.0010)
Distance to the nearest subcenter	−0.0046[a]	−0.0034[b]	−0.0026[c]
	(0.0014)	(0.0017)	(0.0017)
Ruggedness		−0.0010[a]	−0.0009[b]
		(0.0004)	(0.0004)
Distance to coast		0.0015	0.0008
		(0.0016)	(0.0016)
ln(past populations)	N	N	Y
Adjusted R^2	0.21	0.21	0.21

Notes: All regressions include a constant term. 3182 Observations for each regression. Robust standard errors in parentheses.
[a] Significant at the 1 percent level.
[b] Significant at the 5 percent level.
[c] Significant at the 10 percent level.

Table C.2
First-stage and reduced-form results, OLS non-central areas and 1991 distance to the nearest highway.

Dependent variable	First-stage estimates						Reduced-form estimates		
	1991 Dist to ramp			Dist to station			1991–2006 Δln(pop dens)		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Dist to roman road	0.5931[a]	0.5059[a]	0.4609[a]	0.2266[a]	0.1872[a]	0.1916[a]	−0.0012	0.0007	0.0008
	(0.0567)	(0.0473)	(0.0531)	(0.0406)	(0.0379)	(0.0388)	(0.0080)	(0.0081)	(0.0083)
Dist to 19th century road	−0.0085	0.1616[a]	0.2157[a]	0.1723[a]	0.2269[a]	0.2211[a]	−0.0236[a]	−0.0244[a]	−0.0224[a]
	(0.0505)	(0.0486)	(0.0485)	(0.0340)	(0.0328)	(0.0331)	(0.0055)	(0.0061)	(0.0062)
Dist to 19th century rail.	−0.2017[a]	0.0732	0.1060[d]	0.3091[a]	0.3095[a]	0.3029[a]	0.0021	0.0129[c]	0.0134[c]
	(0.0619)	(0.0758)	(0.0725)	(0.0428)	(0.0512)	(0.0509)	(0.0060)	(0.0078)	(0.0079)
ln(1991 pop dens)	Y	Y	Y	Y	Y	Y	Y	Y	Y
Distances	Y	Y	Y	Y	Y	Y	Y	Y	Y
Geography	N	Y	Y	N	Y	Y	N	Y	Y
ln(past pop)	N	N	Y	N	N	Y	N	N	Y
Adjusted R^2	0.52	0.59	0.64	0.56	0.60	0.61	0.30	0.31	0.31

Notes: All regressions include a constant term. 622 Observations for each regression. Robust standard errors in parentheses. Selected instruments are highlighted in gray.
[b]Significant at the 5 percent level.
[a] Significant at the 1 percent level.
[c] Significant at the 10 percent level.
[d] Significant at the 20 percent level.

of Barcelona transportation infrastructure is the Roman roads.[17] Although other ancient roads also existed, Romans were the first to build a sophisticated system with paved and crowned roads. At the beginning of the Roman influence, these main roads were built for military purposes: first, the conquest of Hispania (Spain) and later its defense. These strategic roads passed through mountains and avoided valleys. During the Pax Romana period some of these military roads were abandoned whereas others were modified to pass through less steeper and faster routes. Furthermore, new roads were built in order to improve the accessibility of Hispania. The resulting road system formed a decentralized, mesh-like network that allowed Barcelona (and Hispania) to increase administrative and commercial relations with the rest of the Empire (Font-i Garolera, 1999; Carreras and de Soto, 2010; Bel, 2011).

The second instrument is based on main roads present before the end of the 19th century. This infrastructure was an adaptation of the 18th century postal routes with a subsequent upgrade for automobile use. At the very beginning of the 18th century, the Bourbon dynasty came to power in Spain, succeeding the Habsburg dynasty. The new king, Philip V, changed the Spanish political system from a federation of kingdoms to an absolutist state and, as a result, political power was centralized in the capital. Adopting the Paris model, Madrid became the real "center" of Spain with the construction of a new transportation infrastructure for the postal service funded by the crown (Fig. B.1): a mainly radial network that neglected most of the previous Roman roads. This radial system was designed to improve communications between Madrid and the rest of the new unified king. Since main roads at the end of the 19th century followed the 18th century post routes, I only use the former as instrument.[18]

Finally, I consider the 19th century railroad lines because land transportation and its infrastructure also changed dramatically with the appearance of the steam locomotive. As noted in Appendix A, this network was developed during the second half of the 19th century by private companies such as the Northern Company or the Madrid-Zaragoza-Alicante (MZA) Company and without following any general plan. At the very beginning, railroad development was not regulated and private companies were left to extend the network at their will in search of profits. Subsequent developments were regulated by the 1855, 1867 and 1877 laws, prioritizing projects that connected Madrid with the rest of Spain. As a result, the 19th century railroad was designed following the radial system inherited from the 18th century. See Artola (1978) and Font-i Garolera (1999) for more details on the development of railroads in Spain and Catalonia (Barcelona), respectively.

As outlined earlier, I use vector digital maps based on de Soto (2010) and Carreras and de Soto (2010) to compute instruments. These archaeologists analyze the Spanish economy in different historical periods through the study of transportation infrastructures and related transportation costs and shipping patterns. Their main innovation is the use of spatial data analysis with GIS and the creation of digital maps. As an example of these maps, Fig. B.2 shows Roman roads in the BMR.

Appendix C. Supplemental results

C.1. OLS results

For the entire BMR, Table C.1 presents OLS results for the three extended conditional specifications. Results confirm the positive effect of both railroad and highway improvements on population (density) growth.

Econometrically, the suburbanization process is verified through positive coefficients on distance to the CBD, which indicate that population increases far away from the CBD. Furthermore, negative coefficients on distance to the nearest subcenter suggests that population also increases close to suburban subcenters.

C.2. First-stage results

For non-central areas, columns 1–3 of Table C.2 presents first stage results for the 1991 distance to the nearest highway ramp, and columns 4–6 for the distance to the nearest railroad station. Columns 7–9 shows reduced form results. Each set of regressions includes the three extended conditional specifications that gradually add location variables (columns 1, 4, and 7), geography variables (columns 2, 5, and 8), and history variables (columns 3, 6, and 9).

Following the stragety of Murray (2006a,b), I choose the distance to the nearest 19th century main road as an instrument. Although it does not fulfill both criteria, I choose the distance to the nearest Roman road as the second instrument because each of its first-stage

[17] Although Iberians and Greeks settled in the area before the Romans, the foundation of Barcelona as a city was formalized by Caesar Augustus in 14 B.C. with the name of Iulia Augusta Faventia Paterna Barcino or, its shortened name, Barcino. For more details, see http://en.wikipedia.org/wiki/History_of_Barcelona.
[18] The partial correlation between their distances varies from 55% to 79% for CBD and subcenters samples, respectively.

coefficients are significant. I do not choose distance to the 19th century railroad because its reduced-form results do not support intuition: its positive coefficients indicate that proximity to this infrastructure has a negative effect on growth. For selected instruments, first-stage results indicate that 1991 highways and railroads are close to both Roman roads and 19th century main roads. Reduced-form estimates indicate that proximity to the 19th century main roads has a positive effect on non-central suburban growth.

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